🦉 OWL Agent

PAPER ONLY | Alpaca AI Trading Agents 2026

AUTONOMOUS OPTIONS WHEEL · AI RISK GOVERNOR

An AI that can only say NO.

A deterministic engine sells cash-secured puts and covered calls on hard-coded rails; a Claude layer above it holds exactly two powers — read the regime and veto a new entry. It can never place, size, or force a trade, and it can never block an exit.

+54.6%

selection-first **variant** — 2.5-yr backtest, bot picks 3 stocks/week — vs SPY +45.6%

+32.2%

the **live 5-name engine**'s own 2.5-yr run — live & unattended on Alpaca paper since day 1

100% public

decision journal, backtest artifacts, bugs found & fixed in the open

THE PROBLEM

Options income is a discipline problem disguised as a trading problem

Retail premium sellers must run a boring, repeatable process — the wheel — and can't: the wrong delta, panic buy-backs, one oversized ticker. The usual "AI trading agent" answer makes it worse, because an LLM asked to **pick trades** hallucinates, drifts, and cannot be held to a risk mandate.

MANUAL & EMOTIONAL

Discipline breaks exactly when it matters — challenged puts get bought back in fear, winners get oversized in greed.

LLM PICKERS ARE UNSAFE

A model that can trade can churn an account for engagement and invent entries — no way to prove alignment.

THE AUDIENCE IS HUGE

US options volume: **15.2B contracts in 2025**, a sixth straight record, with retail at 30–60% of flow.

THE SOLUTION

Invert the design: the AI governs, the engine trades

A **deterministic wheel engine** does 100% of the trading on hard-coded rails. A **Claude layer** sits above it as a risk governor with exactly two powers — and every refusal is journaled verbatim.

⚙️ Deterministic engine — does everything

- CSP at delta 0.30, DTE 7–35, premium ≥ 0.5% of collateral
- Take-profit at 50% · roll only for credit at delta 0.60
- Assignment → covered calls above max(cost, upper Bollinger)
- Marketable-limit orders only, exits always before entries

🧠 Claude governor — may only refuse

- **Reads the regime** each cycle (SPY/QQQ trend, vol percentile, GLD/VIXY/BTC, filtered news) → RISK_ON / NEUTRAL / RISK_OFF
- **Vetoes new entries** — one refusal with a reason kills it
- Can never place, size, or force a trade — never block an exit or buy-back

Incentive alignment by construction: an AI that cannot trade cannot churn your account. A deterministic SPY anchor backs it up — the tighter of the two readers governs.

HOW IT WORKS

One cycle, six steps — every 5–15 minutes while the market is open

The broker is the source of truth at every step, and everything the agent decides is appended to a public JSONL decision journal.

1 · Observe
account & market snapshot

→

2 · Gate
kill-switch · drawdown stop · dual regime readers

→

3 · Decide
per-name wheel logic on hard filters

→

4 · AI veto
Claude reviews each entry — one NO kills it

→

5 · Execute
marketable-limit, paper API

→

6 · Journal
every order & reason, public

The regime scales exposure

RISK_ON → 72% of equity as collateral · NEUTRAL → 36% · RISK_OFF → no new entries. Claude's judgment and a deterministic SPY-intraday anchor both apply — the tighter one wins.

The journal is the product

Equity per cycle, every order with its reason, every regime read with its explanation, every error. On night 1 the journal itself caught two production bugs — both fixed live, in public.

SAFETY

Risk is hard-coded, not vibes

Every gate below is a deterministic check that runs before any order — the same gates are stress-tested live on the dashboard's Risk Lab tab.

CONCENTRATION

5 · 2 · 18%

max underlyings · max per correlated sector · max equity per name

TOTAL COLLATERAL

72% / 36%

of equity as collateral (RISK_ON / NEUTRAL) — the rest is dry powder

DAILY DRAWDOWN STOP

−3%

no new entries after a 3% intraday equity drawdown

MARKET SHOCK ANCHOR

−2% · −4%

SPY intraday tiers: budget halves, then entries stop entirely

EARNINGS BLACKOUT

per-name

no new entries near earnings dates (config-driven)

KILL SWITCH

1 file

touch **agent/KILL** and the agent stops submitting anything

Plus: paper endpoint hard-coded (the code cannot reach live money), quality floors on every entry, exit rules that are never vetoed — de-risking is always allowed.

LIVE RUN

Live and unattended on Alpaca paper since day 1

The full system — engine, governor, gates, journal — has been trading paper money on its own from the first evening of the hackathon. Its own decision journal caught two production bugs on night 1; both were fixed live and committed in the open.

PREMIUM COLLECTED

$1,753

4 fills, net of marks

AUTONOMOUS CYCLES

25 live

39 total incl. dry-runs

REGIME READS

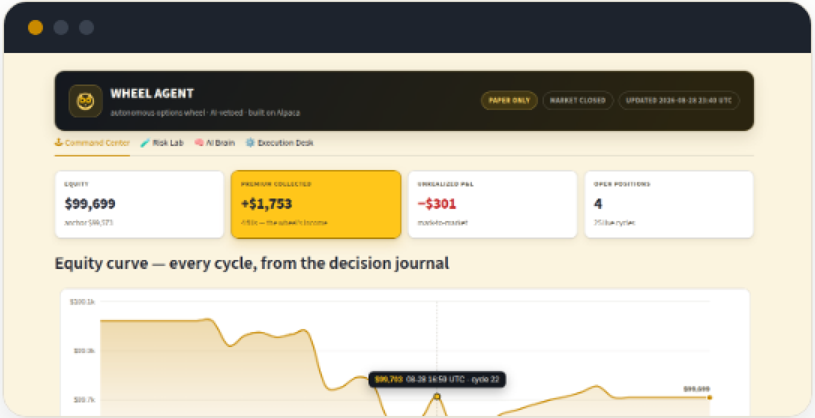
31 / 8 / 0

RISK_ON / NEUTRAL / RISK_OFF

AI VETOES

0

path exercised in dry-run — none needed yet

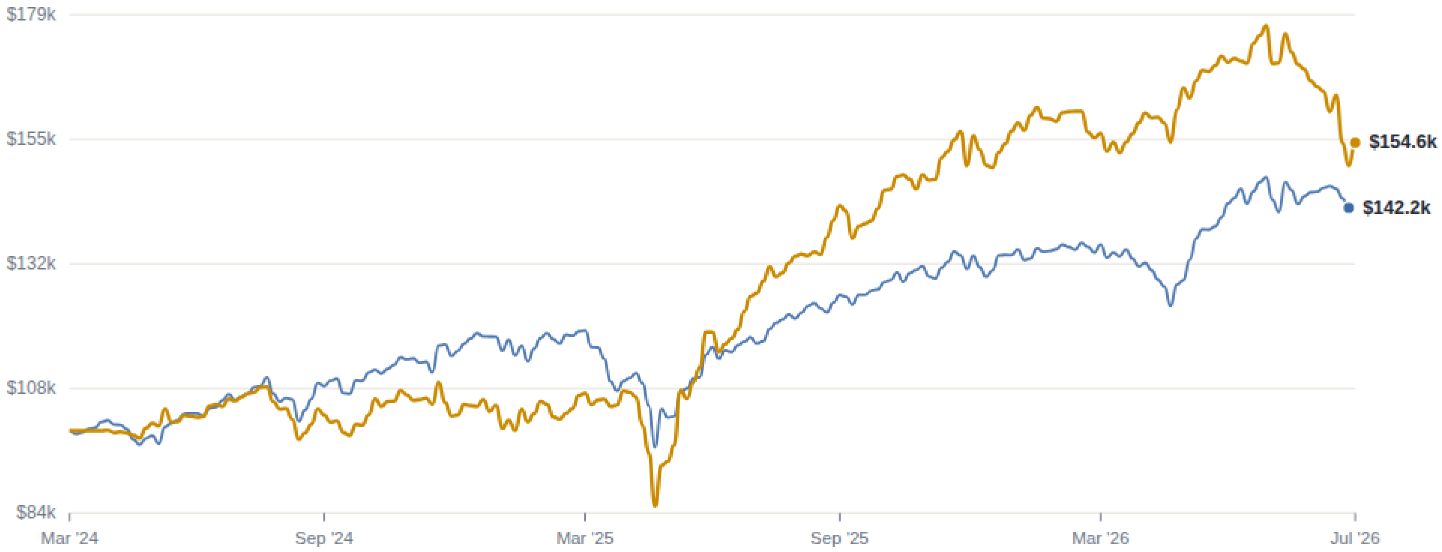


The 6-tab dashboard: Command Center, Risk Lab (live stress test), AI Brain (regime reads & vetoes), Execution Desk, About (backtest methodology), FAQ.

EVIDENCE

Backtested 2.5 years on real option trades — beats SPY in its own bull window

The selection-first mode: each Monday the bot scores a 24-name universe (SMA200 quality gate, 63-day momentum, premium richness) and sells puts on the **top 3 names only**. Every premium is a **real Alpaca option-trade bar**; no parameter was fitted to this window.



EXPLORATIONS — PRE-COMMITTED, HONESTLY REPORTED, FAILURES INCLUDED

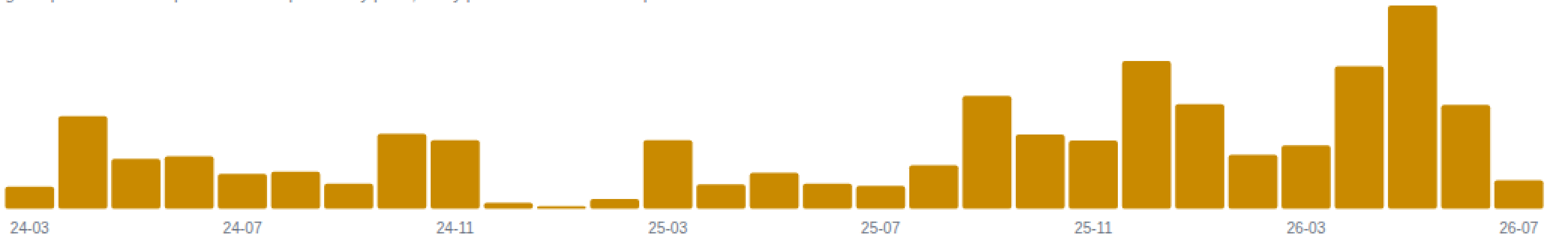
Four configurations, one window, every result published

Each exploration fixes its rules **before** looking at out-of-sample data. We ran a series of backtests to beat SPY buy-and-hold — and picked the top-3 mode because it's **the only configuration that beat SPY in BOTH windows**, the 1-year test and the full 2.5 years.

CONFIGURATION	WINDOW	RETURN	MAX DD	VS SPY +45.6%
Live engine — same rules as production	2.5 yr	+32.2%	18.3%	trades upside for drawdown control
+ Defensive levers — quality screen · regime delta · cash yield	2.5 yr	+34.2%	**12.9%**	Calmar 1.51 vs 1.34 ✓
+ Weekly champion — 1 pick/wk, 40% position	1 yr → 2.5 yr	+46% → +35.3%	→ 27.2%	edge didn't survive the full window
+ Top-3 weekly picks — the slide-7 headline	2.5 yr	**+54.6%**	21.5%	beats SPY on return and Calmar ✓

The selection layer adds measurable value: names the bot picked averaged **+117%** vs the universe's +64%. K=1 beat SPY at 1 year (+46% vs +20%) but failed at 2.5 — rejected; K=2/K=5 lost outright; **K=3 won both windows** (+27% vs +20% at 1 yr) — selected. Momentum screens buy tops in mean-reverting years — which is why the failure row stays on this slide.

gross premium written per month — top-3 weekly picks, every premium a real traded option bar



Full paper trails in agent/runs/ (bt-2026-08-29_*, bt-2026-08-30_topk-sweep, bt-2026-08-30_champion-25mo)

BUSINESS CASE

A paid governor in a market that has never been larger

US listed options printed **15.2B contracts in 2025** — the sixth consecutive record — with retail estimated at 30–60% of flow. Every competitor automates *execution*; none sells an AI risk governor whose alignment is provable.

TAM
$4–18B/yr
~10–15M active US options traders at $30–100/mo (Fermi, sources in repo)

SAM
$0.2–1.8B/yr
systematic premium sellers who will connect a broker API

SOM · 3 YR
$1.5–12M ARR
5–20k subscribers at $25–50/mo, solo → small team

COMPETITOR	PRICE	THEY AUTOMATE	THE GAP WE FILL
Option Alpha	$99–149/mo	your rules, executed	No market judgment, no AI governor, no public journal. We bring the wheel determinism + a veto-only Claude layer + bring-your-own-broker (Alpaca first).
Composer	$32–40/mo	stock algos	
Option Samurai	$39–49/mo	finding trades	
tastytrade	$0 + $1/ctr	education & brokerage	

Revenue: $29/mo core (one broker, full engine + governor + dashboard) · $79/mo pro (multi-account, spreads sleeve) — no payment-for-order-flow; an AI that can only say NO is also a pricing story.

Sources: Cboe State of the Options Industry 2025 · MEMX · Devexperts · competitor pricing pages (Aug 2026)

WHAT'S NEXT

Built on all three Alpaca surfaces, with a public paper trail

TRADING API

alpaca-py

the engine: orders, OPRA greeks, buying-power clamps

CLI

alpaca 0.0.13

ops & monitoring · doctor before every session · backtest data

MCP SERVER

connected

the broker surface, live inside the AI session that supervises it

Next: a defined-risk spreads sleeve, walk-forward re-validation of every gate, the weekly-champion selection mode behind an opt-in with tighter drawdown limits — and the veto-only governor offered as a safety layer for anyone else's trading agent.

📦 **github.com/guntoken/alpaca-wheel-agent**

📓 **agent/journal.jsonl — the decision journal**

🧪 **agent/runs/ — backtests with full artifacts**

📊 **alpaca-wheel-agent.streamlit.app**

Paper trading only · options involve substantial risk · hypothetical backtests don't guarantee future results · not investment advice.